Sodium

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Identity Sheet

Property	Value
Name	Sodium
Other names	Natrium, Na
Chemical formula	Na^+ (ionic form in biological fluids)
Atomic/molecular weight	22.99 g/mol
CAS number	7440-23-5 (element); 7647-14-5 (NaCl)
PubChem CID	5360545
Appearance	Colourless in solution
Electron configuration	[Ne] $3s^1$ (metallic); [Ne] (Na^+ ion)

Structural formula:

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Na+ (monovalent cation, fully hydrated in solution)
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Sodium is the **principal extracellular cation** in the human body, critical for fluid balance, nerve conduction, and muscle contraction. In urine, sodium reflects dietary intake and renal handling, serving as a key diagnostic marker for volume status, AKI aetiology (FENa), and electrolyte disorders.

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
Potassium (K^+)	K^+	39.10	Principal intracellular cation; different renal handling
Chloride (Cl^-)	Cl^-	35.45	Major anion; often measured alongside Na^+; independent regulation
Lithium (Li^+)	Li^+	6.94	Therapeutic drug for bipolar disorder; similar renal handling to Na^+

Medical Information

Origin

Endogenous Sodium is not synthesised endogenously. Total body sodium: ~60 mmol/kg body weight (~4,200 mmol in a 70 kg adult). About 40% is in bone (partially exchangeable), 55% in ECF, 5% in ICF.

Exogenous Dietary sodium intake is the sole source: primarily **sodium chloride (NaCl)** in food. Typical Western diet: 100–250 mmol/day (2.3–5.8 g Na/day or 6–15 g NaCl/day). WHO recommends <2 g Na/day (<5 g NaCl/day). Sources: processed foods, added table salt, bread, cheese, cured meats.

Primary & Secondary Biological Roles

Primary role: - **Extracellular fluid osmolality regulation:** Na^+ is the main determinant of ECF osmolality (~275–295 mOsm/kg), controlling water distribution between compartments. - **Nerve conduction and muscle contraction:** Na^+ influx through voltage-gated channels generates action potentials.

Secondary roles: - **Acid-base balance:** Na^+/H^+ exchange in renal tubules contributes to acid excretion. - **Nutrient co-transport:** Na^+-dependent transporters (SGLT for glucose, amino acid transporters) in gut and kidney. - **Blood pressure regulation:** Sodium balance directly affects blood volume and blood pressure.

Catabolism and Elimination Pathway

- Na^+ is freely filtered at the glomerulus (~25,200 mmol/day at GFR 180 L/day).
- **~99%** is reabsorbed: 67% in proximal tubule, 25% in thick ascending loop of Henle (NKCC2), 5% in distal tubule (NCC), 2–3% in collecting duct (ENaC, regulated by aldosterone).
- Final urinary excretion closely matches dietary intake (steady state).
- Minor losses: sweat (10–40 mmol/L), GI tract (<10 mmol/day normally), saliva.

Expression in Humans

Normal Levels

Compartment	Reference Range
Serum sodium	135–145 mmol/L
Urinary sodium (24-h)	40–220 mmol/day (diet-dependent)
Urinary sodium (spot)	20–200 mmol/L (highly variable)
FENa (fractional excretion)	<1% in prerenal AKI; >2% in intrinsic renal AKI

Factors Influencing Levels **Increased urinary Na:** - High dietary salt intake - Diuretic therapy (loop, thiazide) - Salt-wasting nephropathy - Adrenal insufficiency (low aldosterone) - SIADH (dilutional hyponatraemia with high urine Na) - ATN / intrinsic renal failure (FENa >2%) - Post-obstructive diuresis

Decreased urinary Na: - Low salt diet - Volume depletion / dehydration (prerenal state, FENa <1%) - Heart failure, cirrhosis, nephrotic syndrome (avid Na retention) - Hyperaldosteronism - Pre-renal AKI

Associated Pathologies

Condition	Urine Na Pattern	Key Symptoms
Prerenal AKI (dehydration)	UNa <20 mmol/L, FENa <1%	Oliguria, tachycardia, dry mucosa
Intrinsic renal AKI (ATN)	UNa >40 mmol/L, FENa >2%	Oliguria/anuria, rising creatinine
SIADH	UNa >30 mmol/L, dilute serum	Hyponatraemia, confusion, seizures
Hyperaldosteronism	Low UNa despite hypertension	Hypertension, hypokalaemia
Salt-wasting nephropathy	Inappropriately high UNa	Hyponatraemia, polyuria

Presence in Urine

Should it be normally present? Yes — sodium is normally excreted in urine, matching dietary intake. Urinary Na excretion is the primary mechanism for Na balance.

Normal urinary levels: 40–220 mmol/day (24-h); spot values highly variable (20–200 mmol/L).

Form in urine: Na^+ ion — fully dissociated, hydrated monovalent cation. No crystallisation issues.

Pathological significance:

Urinary Na	Possible Causes	Prevalence
High (>200 mmol/day)	High salt diet, diuretics, salt-wasting, ATN	Common
Low (<20 mmol/L spot)	Volume depletion, CHF, cirrhosis, prerenal AKI	Common

Solubility: NaCl is highly soluble in water (~360 g/L at 20 C). Sodium does not form urinary crystals.

Detection in Urine

Available Clinical Assays

1. **Ion-selective electrode (ISE) — direct or indirect:**
   - **Principle:** Na^+-selective glass or polymer membrane electrode generates a Nernstian potential proportional to Na^+ activity.
   - **Detection:** Potentiometric (voltage vs reference electrode).
   - **LOD:** ~1 mmol/L.
   - **Advantages:** Standard on all automated analysers; rapid (<30 s); precise (CV <2%).
   - **Disadvantages:** Pseudohyponatraemia with indirect ISE in hyperlipidaemic/hyperproteinaemic samples (not relevant for urine).
2. **Flame photometry (historical reference):**
   - **Principle:** Na atoms in a flame emit light at **589 nm** (sodium D line, characteristic yellow emission).
   - **Detection:** Emission intensity at 589 nm proportional to Na concentration.
   - **LOD:** ~0.1 mmol/L.
   - **Advantages:** Simple, robust, multi-element (Na, K, Li simultaneously).
   - **Disadvantages:** Largely replaced by ISE; requires gas flame.
3. **Flame atomic absorption spectroscopy (FAAS):**
   - **Principle:** Ground-state Na atoms absorb 589 nm light from a sodium hollow-cathode lamp.
   - **Detection:** Absorption at **589 nm**.
   - **LOD:** ~0.01 mmol/L.
   - **Advantages:** High sensitivity; reference method.
   - **Disadvantages:** Specialised equipment.

Optimal Urine Type for Measurement

- **24-hour urine** is preferred for assessing total daily Na excretion (dietary intake estimation, Na balance studies).
- **Spot urine** with creatinine correction (UNa/UCr ratio) is used for FENa calculation and acute clinical decisions.
- First morning urine acceptable for spot assessment.
- No special preservatives needed. Stable at room temperature for days.

Actual Gold Standard

Ion-selective electrode (ISE) on automated chemistry platforms (Roche Cobas, Siemens Advia, Abbott Architect) is the universal clinical method. Direct ISE is preferred for urine. CV <2%. Reference method: flame emission photometry at 589 nm.

Interferences in Measurement

Interference	Effect	Mechanism
Lipaemia/hyperproteinaemia	Pseudohyponatraemia (indirect ISE only)	Excluded volume effect; not relevant for direct ISE or urine
Very high K^+	Minor Na^+ ISE interference	Cross-selectivity of membrane
Surfactant contamination	Variable	Alters electrode membrane response
Lithium (high levels)	Minor positive bias	Some ISE cross-react with Li^+

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- **Flame emission (589 nm):** Classic Na detection. The sodium D-line at **589.0 and 589.6 nm** is one of the most intense atomic emission lines.
  - LOD: ~0.01 mmol/L.
  - Sample prep: Dilution in deionised water.
- **UV-Vis:** Na^+ has no absorption in UV-Vis as a simple ion. No direct molecular spectroscopy.
- **NIR:** No useful Na^+ absorption.

Fluorescence Detection

- Na^+ itself is non-fluorescent. Indirect detection via:
- **Sodium-sensitive fluorescent probes:**
  - **SBFI (sodium-binding benzofuran isophthalate):** Ratiometric probe; Ex 340/380 nm, Em 505 nm. LOD: ~0.1 mmol/L in biological fluids. Originally for intracellular Na imaging.
  - **CoroNa Green:** Ex 492 / Em 516 nm. LOD: ~1 mmol/L. Simpler, single-wavelength.
- **Sample prep:** Mix urine with probe solution; calibrate with Na standards.
- **Limitation:** Dynamic range may not cover full urinary Na range (20–200 mmol/L) without dilution.

Raman Detection Na^+ has no Raman-active vibrational modes (monatomic ion). Raman spectroscopy cannot detect sodium.

FTIR Detection Na^+ has no infrared absorption (no molecular bonds to vibrate). FTIR cannot detect ionic sodium. Sodium salts (e.g., NaCl) show lattice modes only in far-IR (<200 cm^{-1}) in crystalline state, not useful for solution analysis.

Voltammetry Detection

- **Potentiometric ISE:** The standard electrochemical method (see clinical assays above). Na^+-selective glass or polymer (bis(12-crown-4) based) membrane.
  - LOD: ~1 mmol/L.
  - Response time: <30 s.
- **Solid-state ISE on chip:** Screen-printed Na^+ ISE for point-of-care. LOD: ~0.5 mmol/L.
- Na^+ cannot be detected by voltammetric methods (CV, DPV) because its reduction potential is too negative (E0 = -2.71 V vs SHE) for aqueous media.

Other Detection Technologies

1. **Inductively coupled plasma optical emission spectroscopy (ICP-OES):** Multi-element analysis at 589 nm. LOD: ~0.001 mmol/L. Reference method for research.
2. **ICP-mass spectrometry (ICP-MS):** Ultimate sensitivity (LOD ~0.0001 mmol/L). Used for multi-element urinary panels.
3. **Paper-based ISE sensors:** Disposable Na^+ sensors on paper substrates with ionophore membranes. LOD: ~1 mmol/L.
4. **Smartphone-based flame photometry:** Miniaturised flame emission with phone camera detection. Proof-of-concept stage.

Summary Table: Detection Methods Comparison

Method	LOD	Key Parameter	Sample Prep	Strengths	Limitations
ISE (direct)	~1 mmol/L	Potentiometric	None/dilution	Standard, fast, precise	Electrode maintenance
Flame emission	~0.01 mmol/L	589 nm emission	Dilution	Reference, multi-element	Gas flame needed
FAAS	~0.01 mmol/L	589 nm absorption	Dilution	High sensitivity	Specialised
SBFI fluorescence	~0.1 mmol/L	Ex 340-380 / Em 505 nm	Probe addition	Imaging capable	Expensive probe
CoroNa Green	~1 mmol/L	Ex 492 / Em 516 nm	Probe addition	Simple	Limited range
ICP-OES	~0.001 mmol/L	589 nm emission	Dilution	Multi-element	Instrument cost
Paper ISE	~1 mmol/L	Potentiometric	None	Disposable, POC	Prototype
Raman / FTIR	N/A	N/A	N/A	Not applicable	No signal

References

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